Rahul Muthu

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Education

Stony Brook University Stony Brook, NY
B.S. in Computer Science and Applied Mathematics Aug 2023 – Dec 2027 (expected)
Relevant Coursework: Data Structures, Systems Fundamentals I & II, Software Development, Operating Systems, Cloud Computing, Data Mining, Theory of Computation, Computational Geometry, Probability Theory.
In progress: Computer Networks, Software Engineering.

Experience

Stony Brook University Stony Brook, NY
Teaching Assistant, CSE 320: Systems Fundamentals II Jan 2026 – Present

- Designed a coverage-guided fuzzer assignment and wrote the reference implementation in C. Instrumented targets report edge coverage through shared memory.
- The fuzzer coordinates parallel target processes with fork/exec, pipes, and signals; a Unix-domain-socket validator exercises crashes, timeouts, and signal handling.
- Wrote autograder and exam questions and led weekly office hours on debugging, memory, processes, and concurrency.

Stony Brook University Stony Brook, NY
Research Assistant Oct 2024 – Aug 2025

- Built PyTorch classifiers for cell-type annotation and evaluated them with grid search and k-fold cross-validation on 27,000 scRNA-seq profiles.
- A label hierarchy and class-balanced loss improved rare-cell-type F1 by 15 points; feature selection cut the input from 23,000 genes to 3,500.
- Moved training to a SLURM pipeline on eight A100 GPUs, reducing epoch time from 130 to 45 minutes; presented the work at the 2025 URECA symposium.

Giesecke+Devrient Dubai, UAE
Software Engineering Intern Jul 2024 – Aug 2024

- Built an XGBoost demand-forecasting prototype from checkout data for cash-inventory planning.
- Implemented a Lambda/Kinesis ingestion path for an analytics dashboard; added Jest tests to its TypeScript service and automated releases through GitLab CI.

Projects

MicroBook
Personal project, single-symbol limit order book in C++20 Feb 2026 – Jul 2026

- The book uses ordered price maps, FIFO lists at each price, and an order-ID index to avoid full-book scans on cancel and modify.
- Matching follows price-time priority at the resting order's price, including partial fills and deterministic trade ordering.
- Strict text-log replay and fixed-seed randomized traces check invariants after each operation; repeat runs compare final snapshots and hashes.

Awards & Leadership

- **UAE IMO National Qualifier**: 6th nationally; invited to the IMO training camp.
- **Stony Brook ICPC Qualifier**: 2nd overall in the junior category.
- **Founder and President, Stony Brook Quant Club**: 80+ members; organized a Jane Street Estimathon and Flow Traders events.

Technical Skills

Languages: Python, C, C++, TypeScript, Java, SQL, Bash **Tools:** Linux, Git, CMake, SLURM, AWS